

RAG Fundamentals

Retrieval-Augmented Generation, usually called RAG, is an approach that allows a language model to retrieve relevant information from an external knowledge source before generating an answer. The model does not need to memorize every fact in the document collection. Instead, it receives only the most relevant passages for the current question.

A basic RAG pipeline begins by splitting documents into smaller text passages called chunks. Each chunk is converted into an embedding. An embedding is a numerical vector that represents the semantic meaning of the text. These vectors can be stored together with the original chunk text in a local database.

When a user asks a question, the system creates an embedding for the question as well. The query embedding is compared with the stored document embeddings. A similarity metric such as cosine similarity can then be used to rank the chunks from most relevant to least relevant.

The highest-ranked chunks are passed to the language model together with the user question. The model is instructed to answer only from the retrieved sources. If the available sources do not contain enough information, the system should state that the information is not available instead of inventing an answer.

In a local RAG system, the documents, embeddings, retrieval process, and answer model can all remain on the user's computer. This makes it possible to build an offline question-answering assistant for a private document collection.